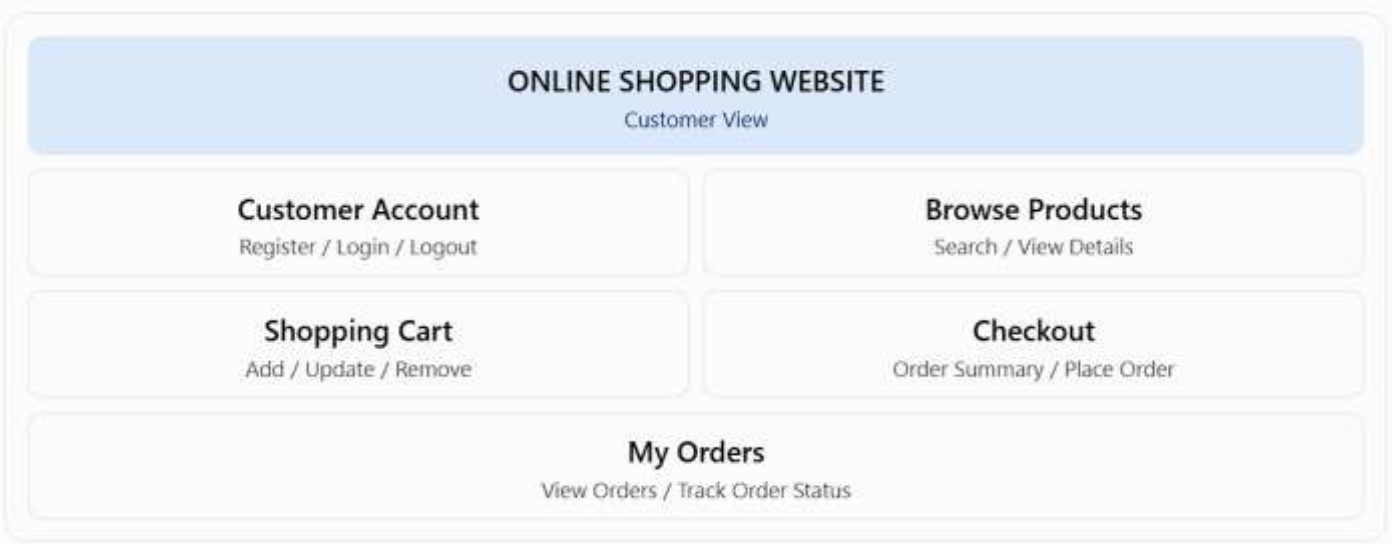
COLLEGE OF ARTS AND SCIENCES
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
1st Semesterm AY 2026-2027
Final Exam in Prof Elective 5 - Advanced Web Development: Back-end
November 2026

II. APPLICATION. Create a dynamic and database-driven online shopping website for customer view using the LARAVEL PHP Framework. The application must allow customers to register and log in, browse available products, manage their shopping cart, place orders, and view their order history and status. (50 points)



IMPLEMENTATION RULES AND REQUIREMENTS

1. API Integration: Implement Laravel API endpoints for communicating between the customer interface and the back-end application. Use appropriate HTTP methods, JSON responses, validation, and authentication.
2. Database Integration: Use Laravel Query Builder or Eloquent ORM for all database operations. Hardcoded product lists and simulated order records are not allowed.
3. Functional Shopping Cart: Implement adding products, viewing cart contents, updating quantities, and removing products. Cart information must accurately reflect the customer's selections.
4. Order Processing: Ensure that submitted orders and their corresponding order details are properly saved in the database. Apply appropriate validation and database transaction handling.
5. Security, Testing, and Advanced Back-End Features: Implement customer authentication and authorization to protect private records. Demonstrate appropriate software testing, Laravel Queues for asynchronous processing, and functional product search.

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Shopping website (Admin View) 60pts

Criteria	Very Good (10–8 pts)	Good (7–5 pts)	Needs Improvement (4–1 pts)
1. 1. Customer Account Module (10 pts)	Successfully implements customer registration, login, and logout. Credentials are validated and authenticated against the database. API integration and appropriate authentication are implemented.	Implements the required account functions, but some features have minor errors or incomplete API integration and validation.	Account functions are mostly nonfunctional, improperly authenticated, or not properly connected to the database.
2. 2. Product Browsing Module (10 pts)	Successfully displays products retrieved using Query Builder or Eloquent ORM. Implements product searching and viewing of product details through functional API endpoints.	Displays products from the database, but search, product details, or API functionality has minor issues or incomplete implementation.	Product browsing is mostly nonfunctional, uses hardcoded product information, or is not properly connected to the database.
3. 3. Shopping Cart Module (10 pts)	Successfully implements adding, viewing, updating, and removing cart items. Cart operations work correctly using appropriate API endpoints and database or session management.	Implements most shopping cart functions, but some operations have errors or incomplete API/database integration.	Shopping cart functions are mostly nonfunctional or do not properly reflect the customer's selected products and quantities.
4. 4. Checkout and Order Processing Module (10 pts)	Successfully implements checkout, accurate order computation, and order submission. Saves order and order details using Eloquent or Query Builder with proper transaction handling. Implements asynchronous order confirmation or notification using Laravel Queues.	Implements checkout and order saving, but has minor issues in computation, database transactions, or asynchronous processing.	Checkout or order processing is mostly nonfunctional. Orders are not properly saved, or order details are incomplete or inconsistent.
5. 5. Customer Order Management Module (10 pts)	Successfully displays customer order history, order details, and status. Uses appropriate database queries and API endpoints, enforces customer data privacy, and demonstrates appropriate testing of application functionality.	Displays order history and status, but has minor issues in data retrieval, API implementation, privacy controls, or testing.	Order management is mostly nonfunctional, customer records are not properly protected, or testing and database integration are insufficient.